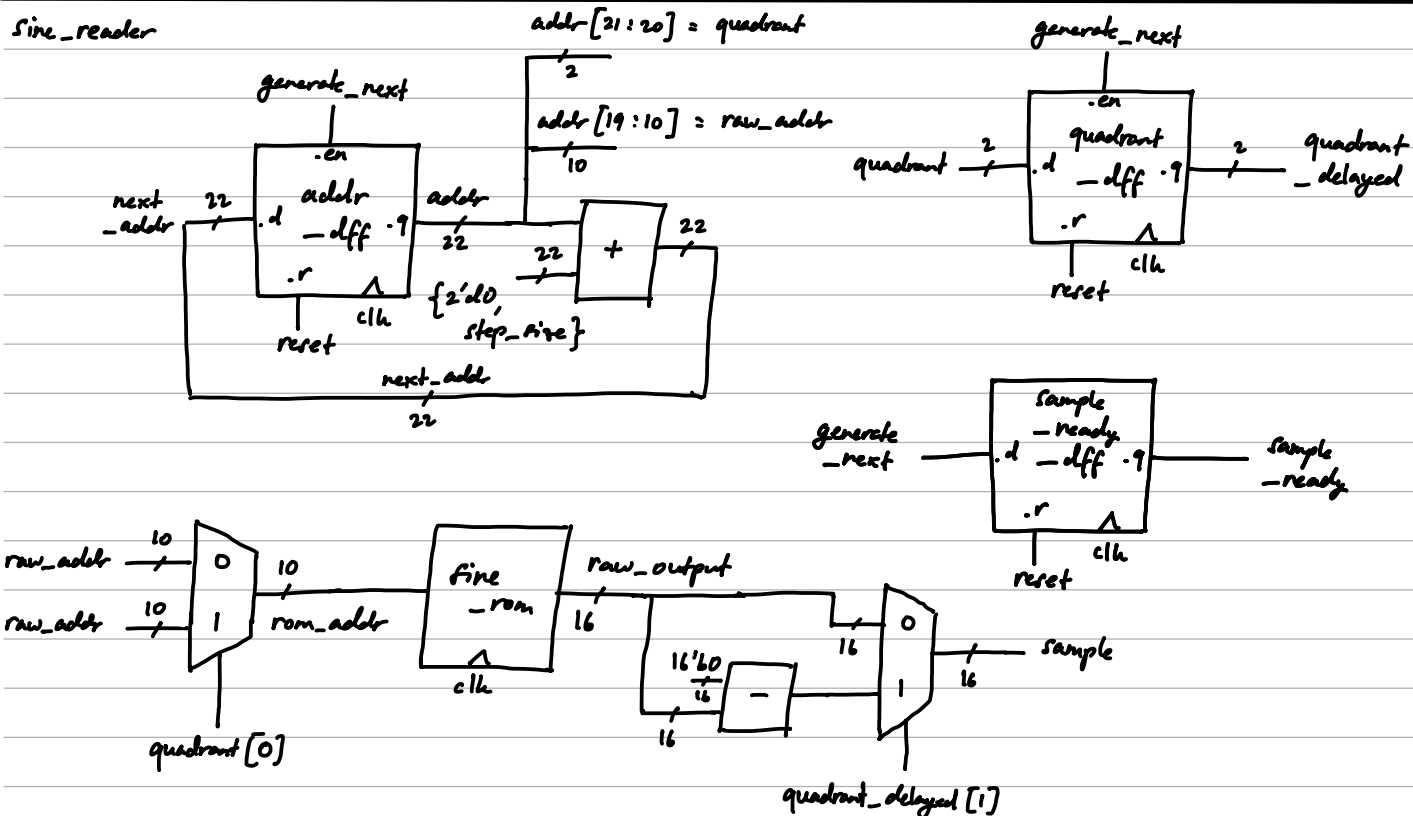
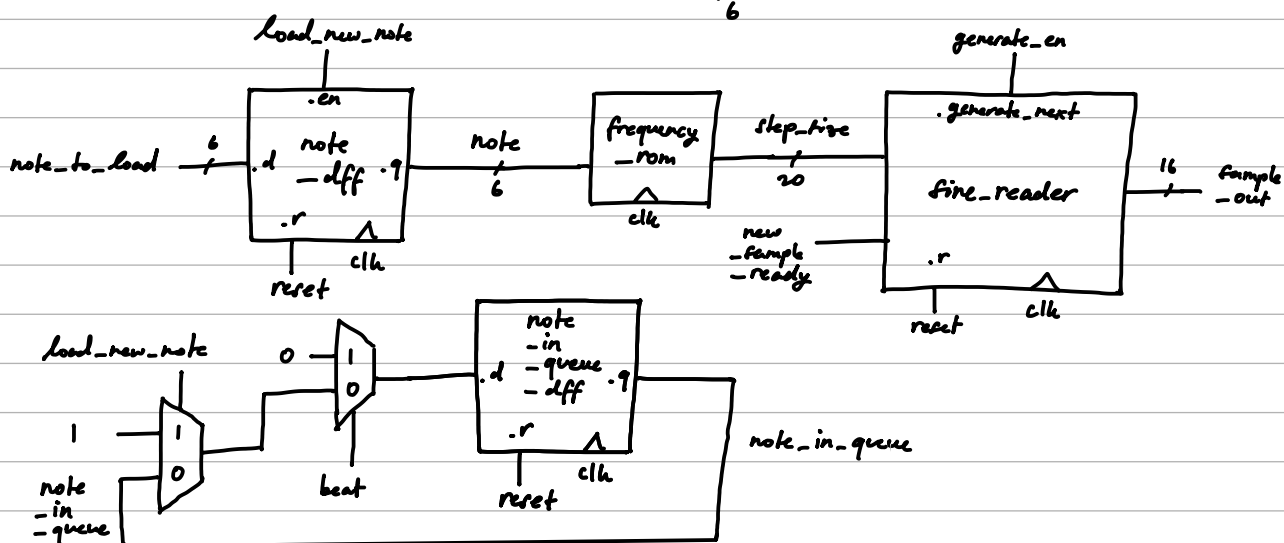
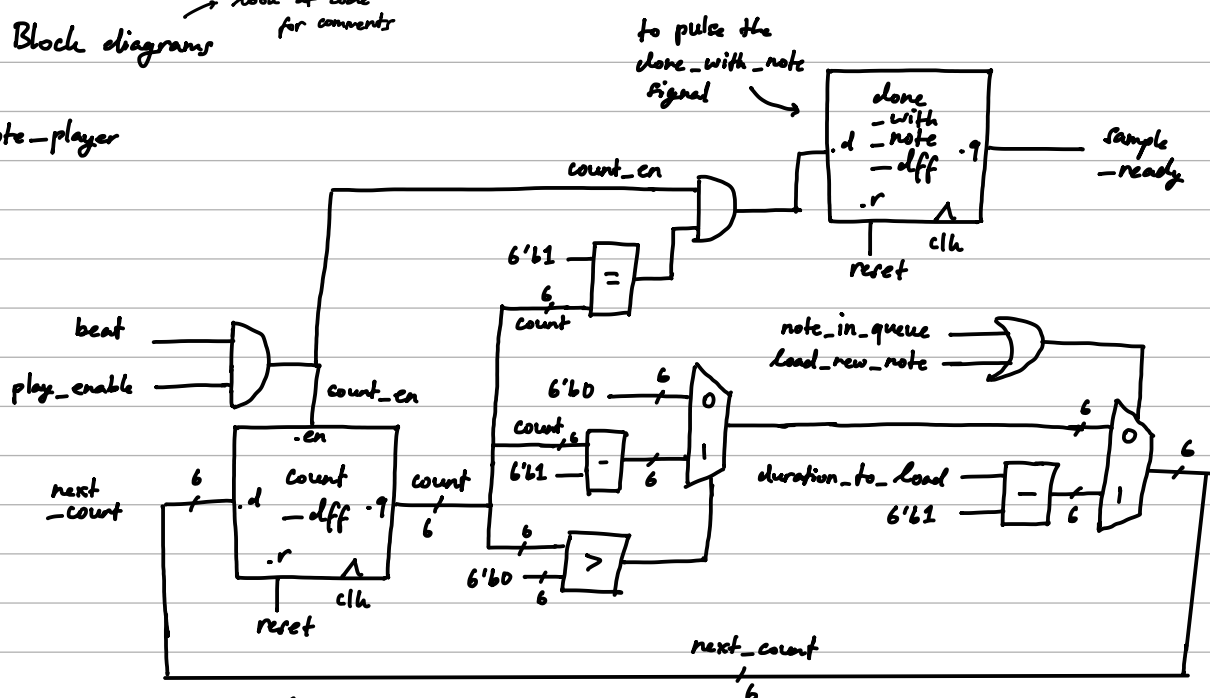


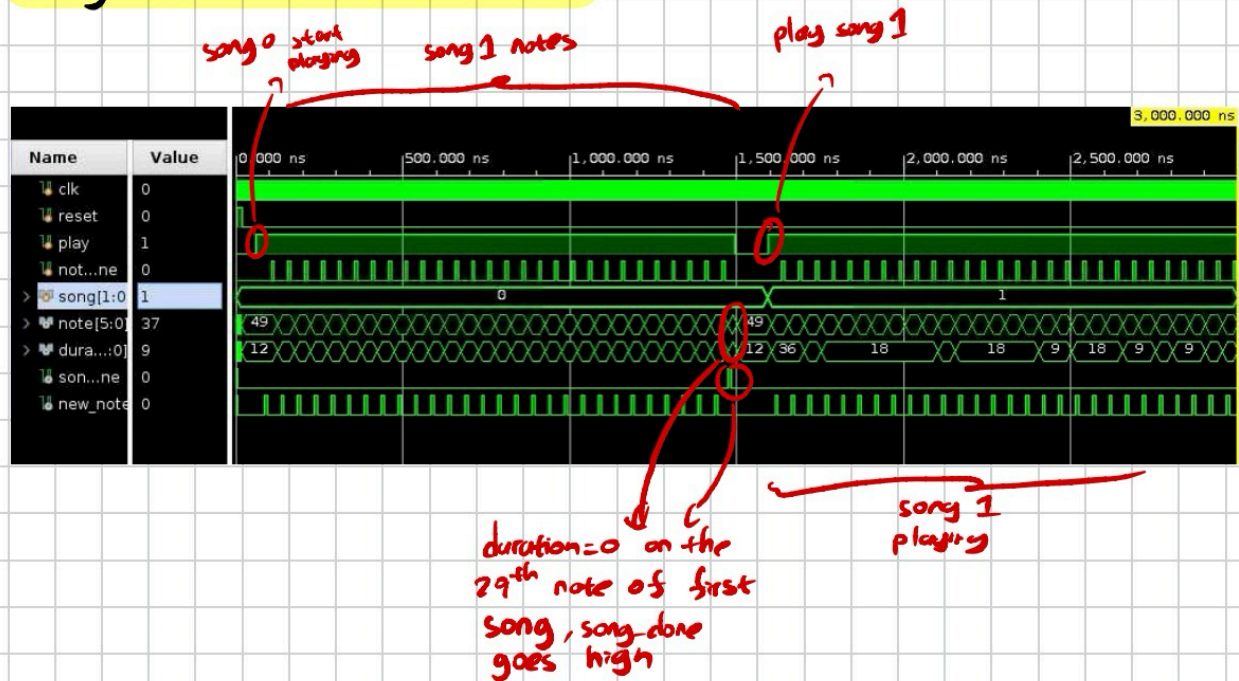
1. Block diagrams ↗ look at code for comments

note-player



2. Annotated testbenches

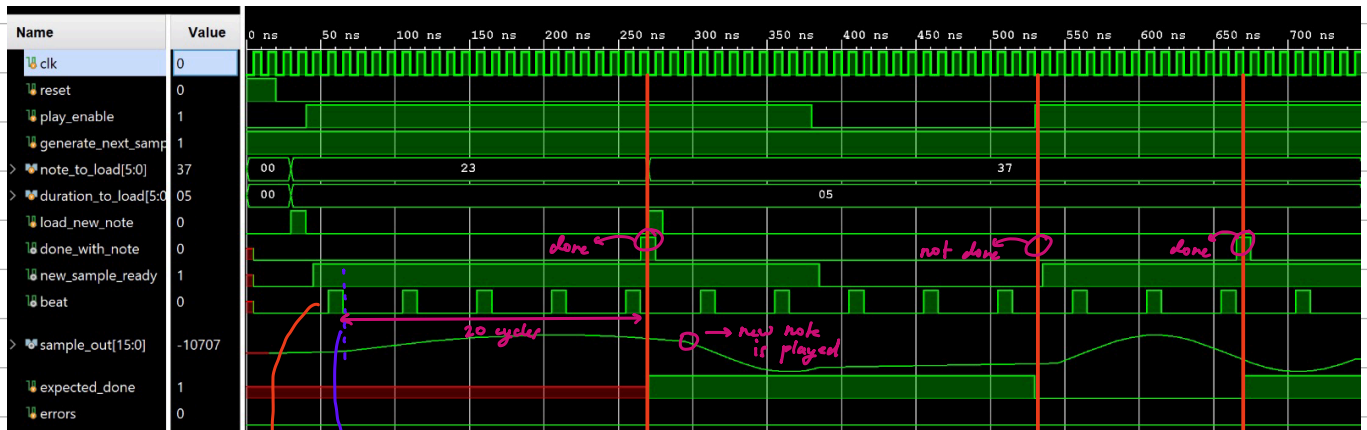
A. Song Reader Testbench



Here is the printed display from the testbench

```
# run 3000ns
Playing note 0 of song 0 : pitch=49 duration=12 time=85000
Playing note 1 of song 0 : pitch=1 duration=8 time=135000
Playing note 2 of song 0 : pitch=51 duration=12 time=185000
Playing note 3 of song 0 : pitch=3 duration=8 time=235000
Playing note 4 of song 0 : pitch=52 duration=12 time=285000
Playing note 5 of song 0 : pitch=4 duration=8 time=335000
Playing note 6 of song 0 : pitch=54 duration=12 time=385000
Playing note 7 of song 0 : pitch=6 duration=8 time=435000
Playing note 8 of song 0 : pitch=56 duration=12 time=485000
Playing note 9 of song 0 : pitch=8 duration=8 time=535000
Playing note 10 of song 0 : pitch=57 duration=12 time=585000
Playing note 11 of song 0 : pitch=9 duration=8 time=635000
Playing note 12 of song 0 : pitch=59 duration=12 time=685000
Playing note 13 of song 0 : pitch=11 duration=8 time=735000
Playing note 14 of song 0 : pitch=13 duration=12 time=785000
Playing note 15 of song 0 : pitch=25 duration=8 time=835000
Playing note 16 of song 0 : pitch=15 duration=12 time=885000
Playing note 17 of song 0 : pitch=27 duration=8 time=935000
Playing note 18 of song 0 : pitch=16 duration=12 time=985000
Playing note 19 of song 0 : pitch=28 duration=8 time=1035000
Playing note 20 of song 0 : pitch=18 duration=12 time=1085000
Playing note 21 of song 0 : pitch=30 duration=8 time=1135000
Playing note 22 of song 0 : pitch=20 duration=12 time=1185000
Playing note 23 of song 0 : pitch=32 duration=8 time=1235000
Playing note 24 of song 0 : pitch=21 duration=12 time=1285000
Playing note 25 of song 0 : pitch=33 duration=8 time=1335000
Playing note 26 of song 0 : pitch=23 duration=12 time=1385000
Playing note 27 of song 0 : pitch=35 duration=8 time=1435000
Playing note 0 of song 1 : pitch=35 duration=36 time=1615000
Playing note 1 of song 1 : pitch=42 duration=36 time=1665000
Playing note 2 of song 1 : pitch=38 duration=54 time=1715000
Playing note 3 of song 1 : pitch=37 duration=18 time=1765000
Playing note 4 of song 1 : pitch=35 duration=18 time=1815000
Playing note 5 of song 1 : pitch=38 duration=18 time=1865000
```

B. note-player_tb



beats every
5 cycles

start of
note

test 1: play note with duration 5
→ completes in 20 cycles as expected

test 2: pause note
→ if the note-player
kept playing, it
should've been completed
but since it paused,
the note isn't done

test 3: unpause
→ when unparsed, it
resumes from the
same place and note
ends as expected

C. fine_reader_tb

step_size = 128.25

1. quadrant 1, floating point addition rounded down

take two steps, expected_addr = 256(.50)

sample = 12539, expected = 12539

2. quadrant 1, floating point addition results in an extra address shift

take two steps, expected_addr = 256.50 + 256.50 = 513(.00)

sample = 23205, expected = 23205

3. quadrant 2, sample is flipped horizontally

take four steps, expected_addr = 513.00 + 513.00 = 1026(.00) → 2(.00)

since this is quadrant 2, flip horizontally

expected_addr = 1023 - 2.00 = 1021(.00)

sample = 32767, expected = 32767

4. quadrant 3, sample is flipped vertically

take eight steps, expected_addr = 1026.00 - 1021.00 - 1 = 4(.00)

since this is quadrant 3, flip vertically

sample = -201, expected = -201

5. quadrant 4, sample is flipped horizontally and vertically

take eight steps, expected_addr = 4.00 + 1026.00 = 1030(.00) → 6(.00)

since this is quadrant 3, flip horizontally and vertically

expected_addr = 1023 - 6.00 = 1017(.00)

sample = -32765, expected = -32765

6. back to quadrant 1

take eight steps, expected_addr = 1026.00 - 1017.00 = 8(.00)

sample = 402, expected = 402

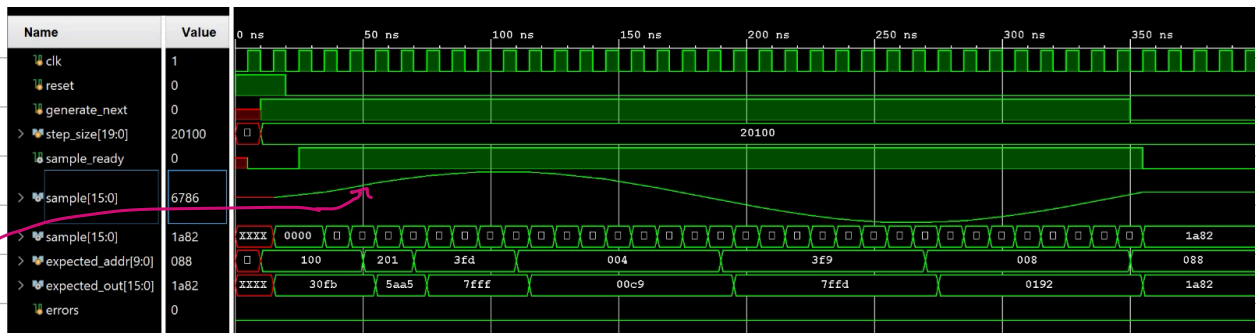
7. off, shift 1 step_size but not played

take eight steps, but it's paused

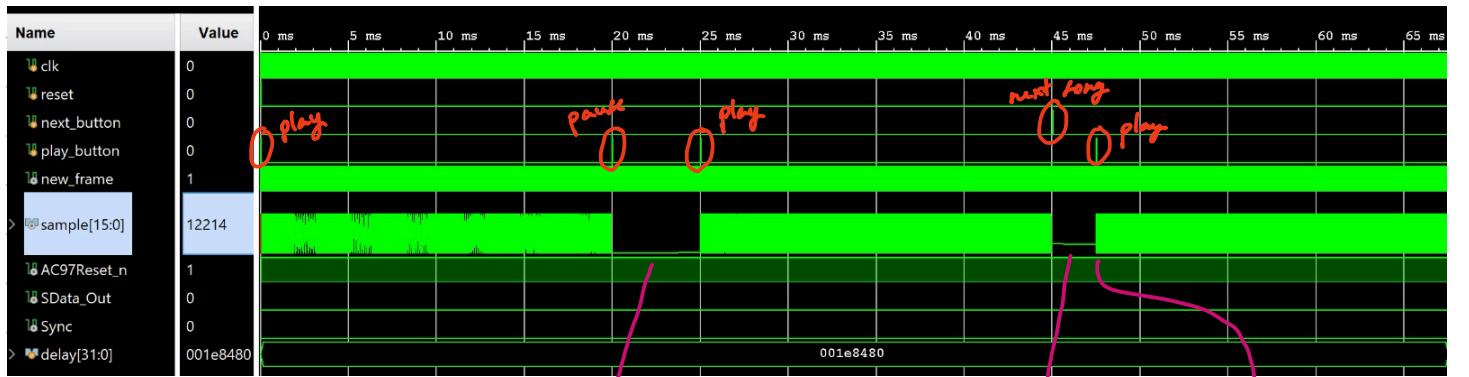
expected_addr = 8.00 + 128.25 = 136(.25)

sample = 6786, expected = 6786

No errors!



D. music_player_tb



console output

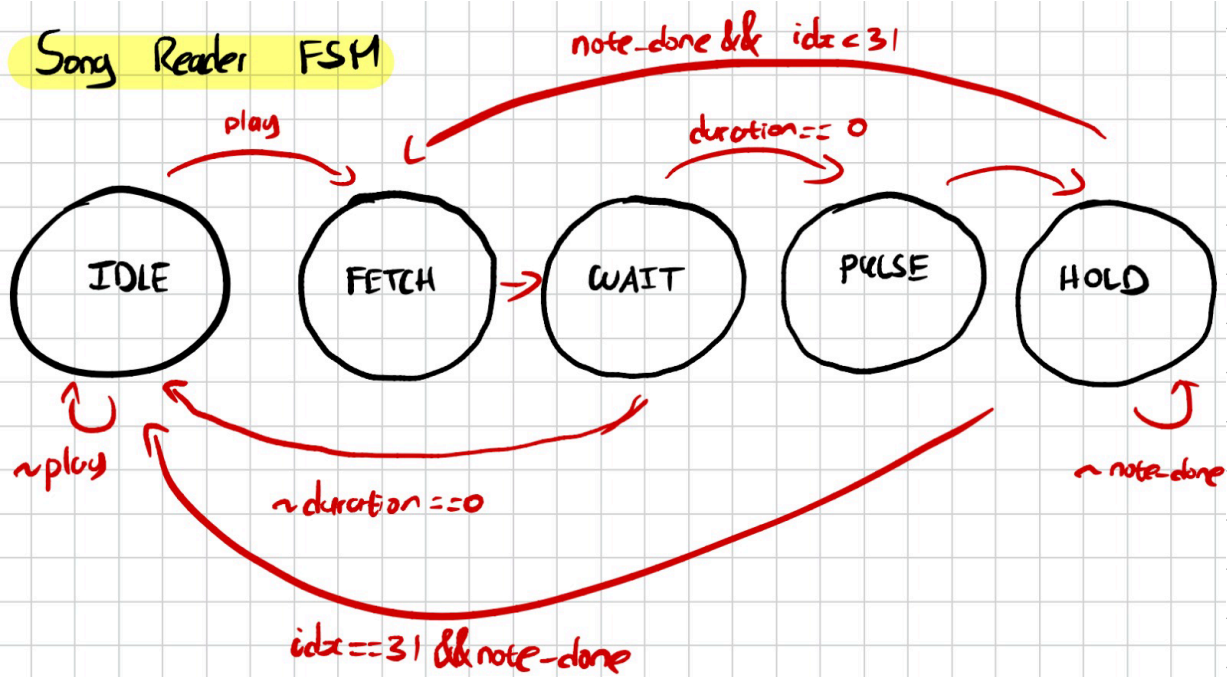
```

Playing note 0 of song 0 : pitch=49 duration=12 time=325000
Playing note 1 of song 0 : pitch=1 duration=8 time=1919905000
Playing note 2 of song 0 : pitch=51 duration=12 time=3199905000
Playing note 3 of song 0 : pitch=3 duration=8 time=5119905000
Playing note 4 of song 0 : pitch=52 duration=12 time=6399905000
Playing note 5 of song 0 : pitch=4 duration=8 time=8319905000
Playing note 6 of song 0 : pitch=54 duration=12 time=9599905000
Playing note 7 of song 0 : pitch=6 duration=8 time=11519905000
Playing note 8 of song 0 : pitch=56 duration=12 time=12799905000
Playing note 9 of song 0 : pitch=8 duration=8 time=14719905000
[15359845000] Audio out: generated 1 seconds.
Playing note 10 of song 0 : pitch=57 duration=12 time=15999905000
Playing note 11 of song 0 : pitch=9 duration=8 time=17919905000
Playing note 12 of song 0 : pitch=59 duration=12 time=19199905000
Pause...
Resume playing song 0...
Playing note 13 of song 0 : pitch=11 duration=8 time=26079905000
Playing note 14 of song 0 : pitch=13 duration=12 time=27359905000
Playing note 15 of song 0 : pitch=25 duration=8 time=29279905000
Playing note 16 of song 0 : pitch=15 duration=12 time=30559905000
[30719845000] Audio out: generated 2 seconds.
Playing note 17 of song 0 : pitch=27 duration=8 time=32479905000
Playing note 18 of song 0 : pitch=16 duration=12 time=33759905000
Playing note 19 of song 0 : pitch=28 duration=8 time=35679905000
Playing note 20 of song 0 : pitch=18 duration=12 time=36959905000
Playing note 21 of song 0 : pitch=30 duration=8 time=38879905000
Playing note 22 of song 0 : pitch=20 duration=12 time=40159905000
Playing note 23 of song 0 : pitch=32 duration=8 time=42079905000
Playing note 24 of song 0 : pitch=21 duration=12 time=43359905000
Next song...
[46079845000] Audio out: generated 3 seconds.
Starting playing song 1...
Playing note 0 of song 1 : pitch=35 duration=36 time=47500405000
Playing note 1 of song 1 : pitch=42 duration=36 time=53119905000
Playing note 2 of song 1 : pitch=38 duration=54 time=58879905000
[61439845000] Audio out: generated 4 seconds.
$finish called at time : 67500380 ns : File "C:/Users/arifk/eel08/eel08-lab4/src/sim/music player tb.v" Line 118

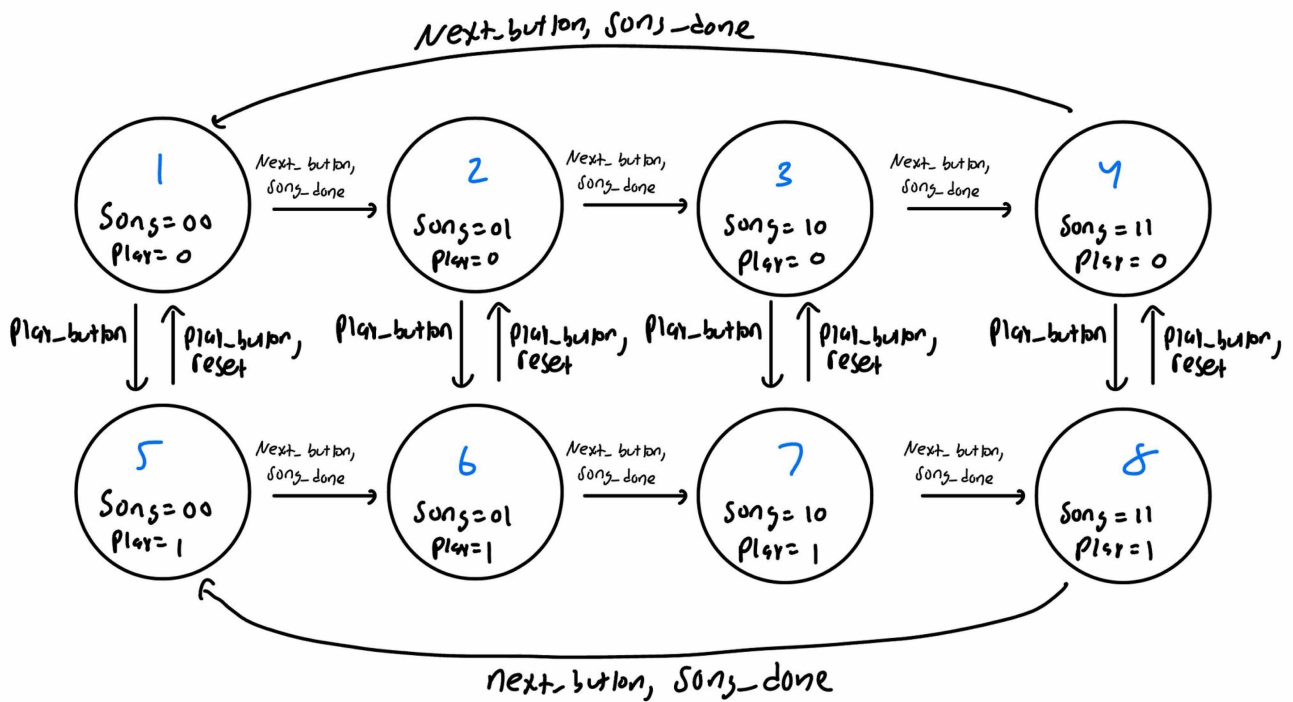
```


3. FSM State Diagram

A. Song Reader FSM



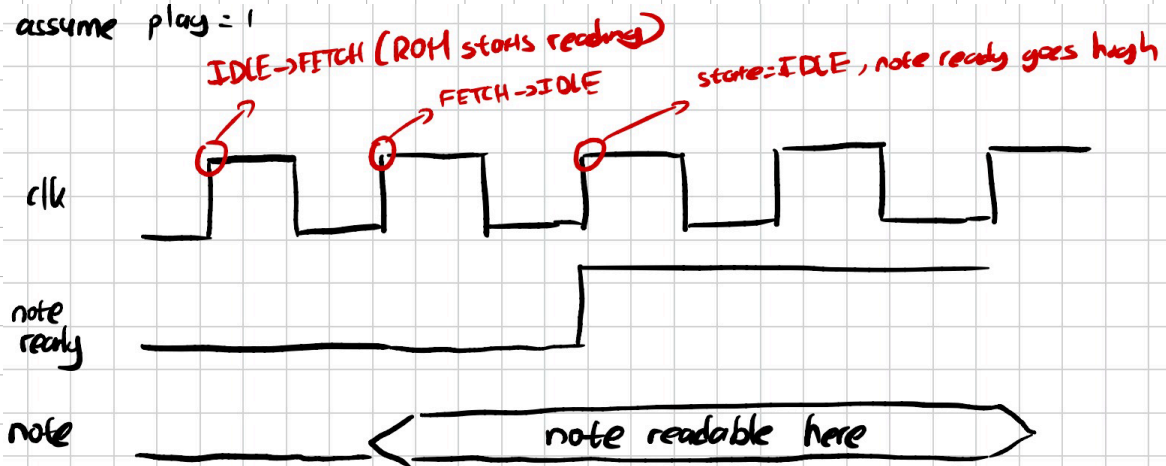
B. MCU



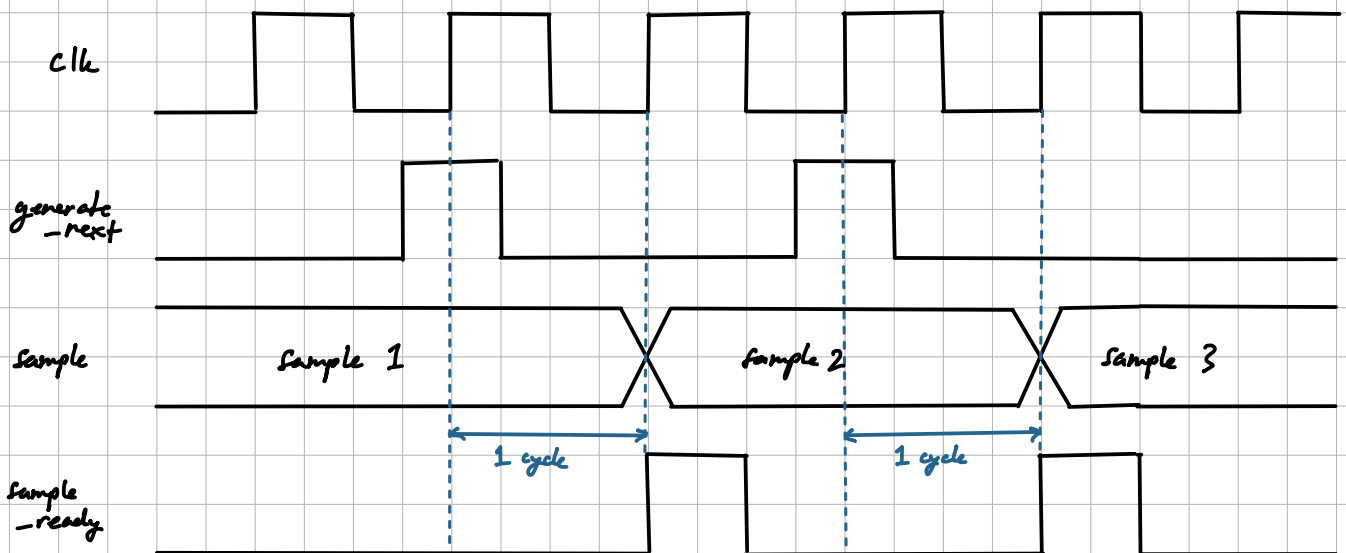
4. Timing Diagram

A. song_reader

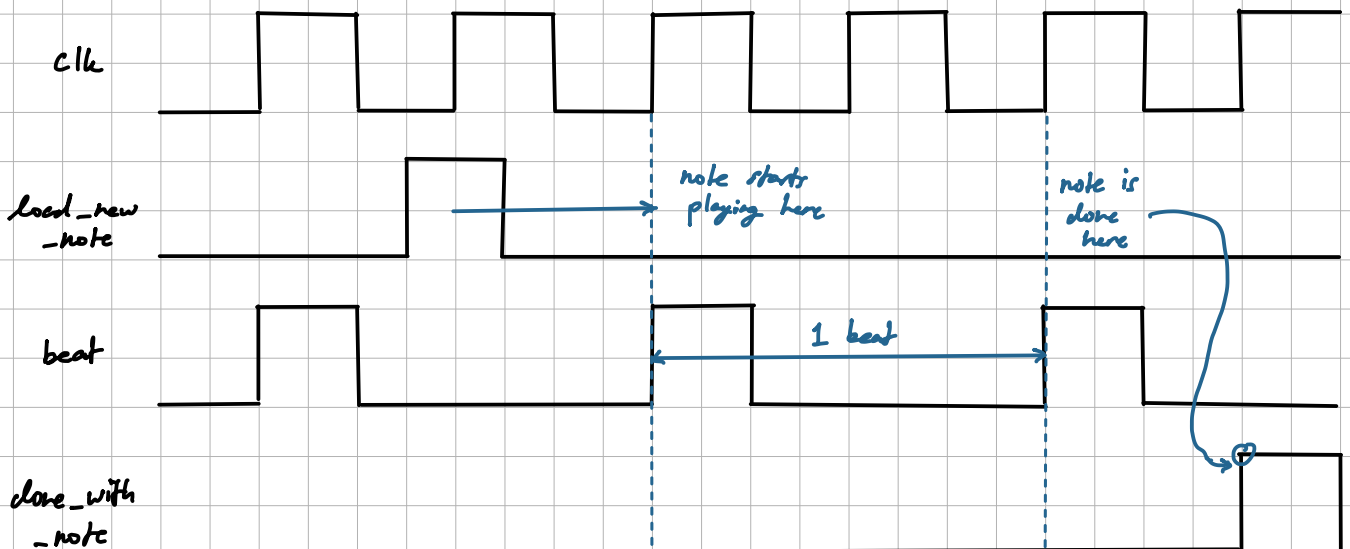
assume $play = 1$



B. fine_reader



C. note_player (loading a duration = 1 note)



5. Timing Report

Path Properties	
Path 1	
Summary	
Name	Path 1
Slack	3.691ns
Source	music_player/note_player/sine_reader/sine_rom/dout_reg/CLKARDCLK (rising edge-triggered cell RAMB18E1 clocked by clk_out1_clk_wiz_0 (rise@0.000ns fall@5.000ns period=10.000ns))
Destination	music_player/pipeline_ff_note_sample/q_reg[15]/D (rising edge-triggered cell FDRE clocked by clk_out1_clk_wiz_0 (rise@0.000ns fall@5.000ns period=10.000ns))
Path Group	clk_out1_clk_wiz_0
Path Type	Setup (Max at Slow Process Corner)
Requirement	10.000ns (clk_out1_clk_wiz_0 rise@10.000ns - clk_out1_clk_wiz_0 rise@0.000ns)
Data Path Delay	6.209ns (logic 3.977ns (64.052%) route 2.232ns (35.948%))
Logic Levels	6 (CARRY4=4 LUT1=1 LUT3=1)
Clock ... Skew	-0.066ns
Clock Uncertainty	0.109ns

→ this path represents the data traveling from the sine ROM to the note_sample. possibly the "flipping the sample based on the quadrant" part?